

Task Breakdown by Role

Feature: Double Opt-In Email Subscription

Repository: Azmith47/public-safety-alerts-aggregator

1. Team Leader / Product Manager

Tasks

- Define MVP scope (subscribe, confirm, unsubscribe, preferences, digest delivery)
- Finalize user stories and acceptance criteria
- Define non-functional requirements (delivery time, reliability, security baseline)
- Align communication language for consent and privacy

Deliverables

- Product requirements document
- Prioritized backlog
- Release checklist

Acceptance Criteria

- All core flows documented and approved
 - Compliance text approved (consent, privacy, unsubscribe wording)
 - Go-live success metrics defined
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2. Backend Developer

Tasks

- Design and implement subscription data model
- Build endpoints:
 - POST /api/subscriptions
 - GET /api/subscriptions/confirm
 - POST /api/subscriptions/resend-confirmation
 - POST /api/subscriptions/unsubscribe
 - POST /api/subscriptions/preferences
- Implement secure token generation and hash storage
- Add status transitions (pending, confirmed, unsubscribed, bounced)
- Implement delivery job (start with daily digest)
- Integrate email provider API

- Add webhook handlers for bounce/complaints

Deliverables

- Database migrations
- API routes and service layer
- Email send pipeline and scheduled job
- Webhook processor

Acceptance Criteria

- Only confirmed users receive alerts
 - Tokens expire and cannot be reused
 - Resend and unsubscribe flows work reliably
 - Bounce addresses are suppressed automatically
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3. Frontend Developer

Tasks

- Build subscription form UI (email + preferences)
- Create confirmation pending page ("Check your inbox")
- Create confirmation success and failure pages
- Build unsubscribe result page
- Build preference management page (token-based access)
- Add user-friendly error states and loading states

Deliverables

- New and updated UI components and pages
- API integration layer
- Validation and UX messaging

Acceptance Criteria

- Full flow is intuitive and works on mobile and desktop
 - No technical or security details leaked in UI messages
 - Invalid and expired token pages provide clear next steps
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4. DevOps Engineer

Tasks

- Configure environment secrets for email provider keys
- Configure sender domain authentication (SPF, DKIM, DMARC)
- Set up scheduled job infrastructure for digest sending
- Configure monitoring and alerting (email failures, webhook failures, queue delays)
- Ensure HTTPS and secure headers in deployed environment

Deliverables

- Secure environment configuration
- Domain mail authentication setup
- Job scheduler configuration
- Monitoring dashboards and alerts

Acceptance Criteria

- Mail authentication passes validation checks
 - Scheduled jobs run reliably
 - Alerts trigger on delivery failure thresholds
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5. Security Engineer

Tasks

- Review token lifecycle and entropy
- Validate secure storage (hashed tokens only)
- Review rate limiting and abuse protections
- Verify anti-enumeration API behavior
- Perform threat modeling (subscription abuse, replay, spam vectors)
- Review logging for security and privacy balance

Deliverables

- Security review report
- Required hardening tasks
- Sign-off checklist

Acceptance Criteria

- No plaintext sensitive tokens stored
 - Abuse protections in place and tested
 - No user enumeration via endpoints
 - Security sign-off completed before production
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6. QA Team

Tasks

- Write test plan for end-to-end subscription lifecycle
- Functional tests for all endpoints and UI paths
- Negative tests (invalid token, expired token, duplicate subscribe, rapid retries)
- Verify only confirmed users receive digests
- Test webhook-driven state changes (bounced/complaints)
- Cross-browser and mobile responsiveness checks

Deliverables

- Test cases and automated regression suite
- Bug reports with reproduction steps
- Final QA sign-off report

Acceptance Criteria

- All critical test cases pass
 - No P1 or P2 defects open at release
 - Regression suite added to CI pipeline
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7. Data / Analytics Team

Tasks

- Define operational metrics:
 - Pending to confirmed conversion rate
 - Bounce rate
 - Complaint rate
 - Unsubscribe rate
 - Delivery latency
- Build lightweight dashboard for team visibility

Deliverables

- KPI definitions
- Dashboard and reporting queries

Acceptance Criteria

- Team can track subscription funnel health weekly
- Alerts exist for abnormal bounce or complaint spikes

8. Suggested Task Timeline

Phase

Focus

Phase 1 Data model, core APIs, basic UI

Phase 2 Email integration, confirm/unsubscribe flows, digest sender

Phase 3 Security hardening, webhook handling, QA regression, release preparation

9. Definition of Done (Team-Wide)

This feature is considered **done** when all of the following are true:

- Double opt-in flow works end-to-end in production
 - Only confirmed users are emailed
 - Unsubscribe works from every email
 - SPF, DKIM, and DMARC configured and validated
 - Rate limiting and anti-enumeration verified
 - QA, Security, and Product Manager sign-offs completed
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Last Updated: August 16, 2026